ModelIt!

Student Activity Pack

What Do Animals Need to Survive?

The Science of Animal Needs

Grade 1 | NGSS 1-LS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Draw a picture of an animal in a place where it has everything it needs. Label the things the animal needs.**

2. **Circle what animals need to survive: food, a TV, water, a phone, air, shelter, toys.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Resources Available

Animal Health

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Plenty of Resources

Resources Available set to high

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Very Few Resources

Resources Available set to low

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What do animals need to stay alive and healthy? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design an Animal Habitat

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YOUR ENGINEERING MISSION

Build a model habitat for a chosen animal that provides everything it needs to survive.

THE SCENARIO

A wildlife rescue center needs your help! They found a lost animal and need to build it a perfect home. Design a habitat that gives the animal food, water, air, and shelter. Make sure it has everything the animal needs!

Engineering Design Thinking

1. WHAT ANIMAL ARE YOU BUILDING A HABITAT FOR?
2. WHERE WILL THE FOOD BE IN YOUR HABITAT?
3. HOW WILL THE ANIMAL GET FRESH WATER?
4. WHAT WILL THE SHELTER LOOK LIKE TO KEEP THE ANIMAL SAFE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wildlife Biologists study animals in their natural habitats to make sure they have everything they need to survive. They earn $45,000-$80,000/year.